Circ.), the average scores of COLE_{SV} with the H_{proxy} model have shown approximately 30%, 91%, and 43% improvements over COLE_{R}. Interestingly, despite the two algorithms demonstrating similar performance on the right-hand starting position of the Asymm. Adv. layout, COLE_{R} exhibits superior performance when initiated from the left-hand starting position.

7.6 Evaluation with Human Players

We recruited 130 students as participants from universities in different majors, and each provided written informed consent. Each participant received a reward of 50 CNY for their participation. To motivate them to be more engaged and attentive, we established a goal (a number of rewards) for each layout. Participants who successfully reached the goal were awarded an additional bonus of 5 CNY for each layout. Experiments were conducted online where each participant completed their own experiment on a certain web page using a computer, and each experiment costs about 15 to 20 minutes. We do not collect any personally identifiable information, and no significant risk to participants was expected.

7.6.1 Experimental Setting

Participants were first acquainted with the experiment and the rules of Overcooked, with details provided in Section 6. Each participant played a sequence of 5 pairs of games, totaling 10 rounds, with 2 agents (one being COLE, while the other was randomly selected from the four agents in section 7.2.1). The experiment employed the in-group setting for each baseline and COLE_{SV} pair. We also randomized the order of agents within each pair to account for skill differences arising from the inner order variance.

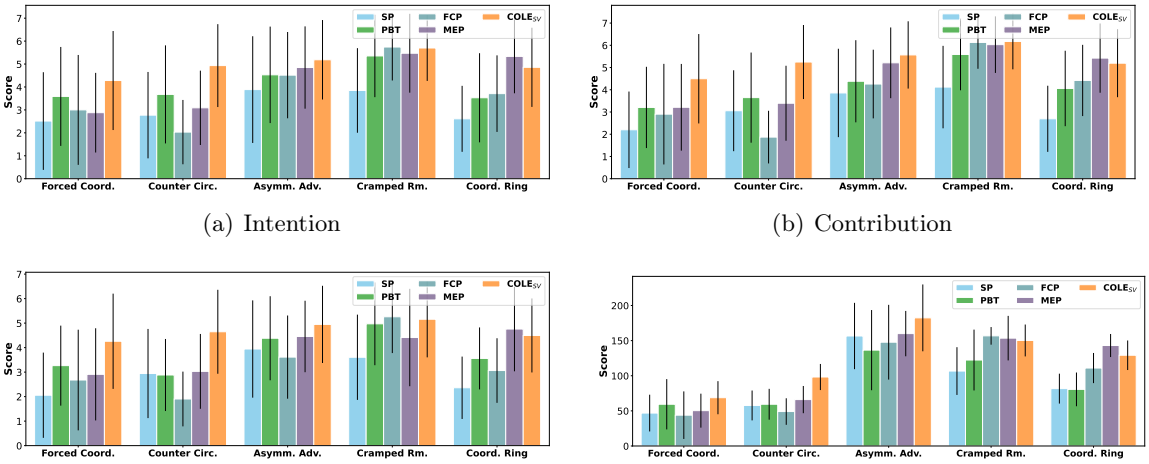
The sequence of five different layouts remained consistent, and we did not compare objective scores across layouts. The five pairs corresponded to five distinct layouts, arranged in the same order for all experiments (1. Cramped Rm., 2. Asymm. Adv., 3. Coord. Ring, 4. Forced Coord., 5. Counter Circ.). Each pair featured one round with the COLE_{SV} agent and one round with the other agent. Under these conditions, we can assume that each participant possesses the same skill level and prior knowledge when encountering the same layouts.

During the experiment, participants were unaware of the specific algorithm names. To ensure fairness, only the color of the chef's hat in the game was used to differentiate between the two algorithms. Each game lasted for 1 minute (approximately 400 steps).

7.6.2 Results of Human Evaluation

In this section, we discuss the findings from our Human-AI experiment. We initially recruited 148 participants from Shanghai Jiaotong University to participate in the study. However, we had to exclude 18 questionnaires from our analysis due to issues such as incomplete responses or negative scoring (e.g., awarding 0 points to all aspects). Thus, our final sample consisted of 130 valid data entries.

The demographic breakdown of the 130 valid participants is as follows: 91 males and 39 females. In terms of prior experience with the Overcooked video game, 70 participants had never played it, 28 had played it but did not consider themselves skilled, 25 regarded themselves as intermediate players, and 7 viewed themselves as experts.



(a) Intention (b) Contribution

(c) Teamwork (d) Rewards

Figure 12: Comparison of subjective evaluation scores and rewards between our COLE$_{\text{SV}}$ method and baseline algorithms across five different layouts. "Intention", "Contribution" and "Teamwork" are shorts for "I understand the agent's intentions", "The agent is contributing to the success of the team", and "the agent and I have good teamwork", respectively. Fig. (d) depicts the average rewards comparison. Our COLE$_{\text{SV}}$ method obviously outperforms the baselines in Forced Coordination, Counter Circulation, and Asymmetric Advantage. The first three layouts necessitate cooperation for higher rewards. In the simplest layout, Cramped Room, PBT, FCP, MEP, and COLE$_{\text{SV}}$ all achieve comparably high scores when playing with humans. In the Coordinated Ring layout, COLE$_{\text{SV}}$'s performance is slightly below MEP but surpasses the other algorithms.

As described in Section 7.2.1, all participants are required to fill in after-game and final questionnaires. The after-game questionnaires are about the subjective evaluation of the playing between participants and one AI agent. The subjective evaluation needs human players to score three questions from 1 to 7, "the agent and I have good teamwork" (short as teamwork), "the agent is contributing to the success of the team" (short as contribution), and "I understand the agent's intentions" (short as intention). As illustrated in Fig. 7.6.2, our proposed COLE$_{\text{SV}}$ surpasses all baseline algorithms in three layouts that require cooperation (Forced Coord., Counter Circ., and Asymm Adv) based on subjective evaluations of intention, contribution, teamwork, and the objective metric of game rewards. Additionally, human players perceive that all methods, except SP, exhibit similar performance in the Crampped Rm layout. In Crampped Rm., both players are situated within a small rectangular area, enabling them to achieve high scores even with minimal cooperation.

In the Coord. Ring layout, COLE$_{\text{SV}}$ marginally underperforms MEP yet surpasses other baseline algorithms. Analyzing human and MEP co-play trajectories revealed MEP's enhanced performance in human subjective experiments arises from its consistent counter-clockwise strategy on the layout. This allows humans to effortlessly adapt to the MEP

Metrics	COLE_{SV} v.s. Baselines			
	SP	**PBT**	**FCP**	**MEP**
Fluency	0.87	0.71	0.78	0.60
Preference	0.88	0.70	0.85	0.68
Understanding	0.80	0.63	0.78	0.61

Table 4: The average scores obtained by COLE_{SV} when participants were asked to answer three evaluation questions comparing COLE_{SV} to an assigned baseline. The questions were: "Which agent cooperates more fluently?" (abbreviated as fluency), "Which agent did you prefer playing with?" (abbreviated as preference), and "Which agent did you understand better?" (abbreviated as understanding). A value closer to 1 indicates a stronger preference for COLE_{SV} among participants.

agent during human-AI experiments by selecting the opposite direction. This predictability leads to misconceptions in questionnaire responses regarding AI's contribution, intention, and teamwork. Participants may perceive a better grasp of MEP's intentions and a more substantial contribution from the agent, but this is primarily due to its easily adaptable and fixed strategy. Nonetheless, when MEP collaborates with diverse partners lacking human-level intelligence, its performance declines, as shown in Fig.9. In contrast, COLE's strategies exhibit greater diversity and are less predictable than MEP's. COLE adapts its routes based on varying situations, which can lead to more conflicts. Consequently, in human subjective evaluations, COLE slightly underperforms compared to MEP. Trajectory visualizations of humans with MEP and COLE are available on our demo page.

In addition to evaluating individual games, human players are also asked to compare the two AI models they played with. They are required to rate the AI performances across three dimensions: fluency through "Which agent cooperates more fluently?", preference by "Which agent did you prefer playing with?", and understanding with "Which agent did you understand better?". The mean rating scores are presented in Table 1.

When the value is greater than 0.5, it indicates that human players prefer COLE_{SV} over the baseline. Consequently, a value closer to 1 signifies a more pronounced preference for COLE_{SV} compared to the baseline model. As demonstrated in the table, all values are larger than 0.6, suggesting that human players concur that COLE_{SV} performs better.

When compared to the SP method, COLE_{SV} receives the highest scores (above 0.8), implying that participants believe that COLE_{SV} will outperform SP with a probability of over 80%. In comparison to the state-of-the-art MEP, COLE_{SV}'s scores also exceed 0.6. These results indicate that human players agree that COLE_{SV} performs better in terms of fluency, preference, and understanding compared to the baseline.

In summary, human players concur that COLE_{SV} is more understandable and contributes significantly to their teamwork, particularly in the three layouts that necessitate cooperation: Forced Coord., Counter Circ., and Asymm Adv.